

Notes on General Equilibrium Theory

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Contents

Chapter 1

Main Results

1.1 Summary of equilibrium notions

Table 1: Walrasian and Financial Market equilibria

Equilibrium	Variables	Object space	Efficiency
Abstract Exchange CE	(x, p)	$\mathbb{R}_+^{LI} \times \mathbb{R}_{++}^L$	1st & 2nd WT
Exchange w/ Production	(x, y, p)	$\mathbb{R}_+^{LI} \times Y \times \mathbb{R}_{++}^L$	1st & 2nd WT
Arrow–Debreu (exchange)	(x, ϕ)	$\mathbb{R}_+^{L(S+1)I} \times \mathbb{R}_{++}^{L(S+1)}$	1st & 2nd WT
Arrow–Debreu w/ Production	(x, y, ϕ)	$\mathbb{R}_+^{L(S+1)I} \times Y \times \mathbb{R}_{++}^{L(S+1)}$	1st & 2nd WT
Financial Market Eq. (FME)	(x, z, p, q)	$\mathbb{R}_+^{L(S+1)I} \times \mathbb{R}^{JI} \times \mathbb{R}_{++}^{L(S+1)} \times \mathbb{R}^J$	Constr. PO if $L = 1$; fails if $L > 1$ (G–P)
FME with No-Arbitrage	(x, π, p)	$\mathbb{R}_+^{L(S+1)I} \times \mathbb{R}_{++}^S \times \mathbb{R}_{++}^{L(S+1)}$	= FME (reformulation)
FME + Production + Rational Conj.	$(x, k, z, \theta, p, q, Q)$	FME dim. $\times K^H \times \mathbb{R}_+^{HI} \times (\mathbb{R}_+^K)^H$	Generically not constr. PO
Representative Agent FME	(x^R, z^R, p, q)	FME w/ $I = 1$; $x^R = \sum_i \omega^i$, $z^R = 0$	Inherits FME; autarchic

Table 2: Dynamic, sunspot, and asymmetric-information equilibria

Equilibrium	Variables	Object space	Efficiency
Sunspot Equilibrium	(x, z, p, q) on $S \times \Sigma$	FME dim., indexed by $\sigma \in \Sigma$	Generically not PO
Sequential FME	$\{(x^i, z^i)_i, q\}$ per node s^t	Indexed by event tree $\mathcal{N}$	Static FME node-by-node
Sequential Arrow–Debreu	(x^*, p^*) per node s^t	$\mathbb{R}_+^{LI \cdot \mathcal{N} } \times \mathbb{R}_{++}^{L \cdot \mathcal{N} }$	1st & 2nd WT
Rational Exp. Eq. (REE)	$\phi : \Sigma^I \rightarrow \mathbb{R}_+^J + \text{alloc.}$	Price-map space $\{\phi : \Sigma^I \rightarrow \mathbb{R}_+^J\}$	Info aggregation; not PO
P–T (Moral Hazard)	$(x, e, q(\cdot))$	$\mathbb{R}_+^S \times E \times \{q : E \rightarrow \mathbb{R}_+^S\} + \text{IC}$	Incentive-constr. PO
P–T (Adverse Selection)	$(x^e)_{e \in E}, q(\cdot)$	$\mathbb{R}_+^{SE} \times \{q : E \rightarrow \mathbb{R}_+^S\} + \text{truth-telling}$	Generically <i>not</i> IC-PO

1.2 Concepts

1.2.1 Boundary condition

The usual boundary condition used in the existence proof is:

$$p^n \rightarrow p \text{ with some } p_l = 0 \Rightarrow \max\{z_1(p^n, w), z_2(p^n, w), \text{dots}, z_L(p^n, w)\} \rightarrow \infty.$$

1.2.2 Aggregate Excess Demand Function Properties

Let $z(p, w) : \mathbb{R}_{++}^L \times \mathbb{R}_{++}^{LI} \rightarrow \mathbb{R}_+^L$ be the excess demand function. It has the following properties (“SWBBH”)

1. smoothness: $z(p, w)$ is C^1
2. Walras Law: $pz(p, w) = 0$.
3. bounded from below: $z(p, w) \geq -s$
4. boundary condition: $p^n \rightarrow p$ with some $p_l = 0$ then, $\max\{z_k(p, w)\} \rightarrow \infty$.
5. homogeneity of degree 0: $z(\lambda p, \lambda w) = \lambda^0 z(p, w) = z(p, w)$.

1.3 Propositions

1.3.1 Sunspots

Intuitively, a sunspot equilibrium means that we can have different equilibrium outcomes even when economic fundamentals are the same.

$\exists$ non-trivial sunspot equilibrium $\Rightarrow$ equilibrium indeterminacy or multiplicity.

Cass and Shell. When markets are complete, sunspot equilibria do not exist. That is, if

$$\omega_{s,e} = \omega_{s,e'} = \omega_s,$$

then

$$x_{s,e} = x_{s,e'}.$$

But when markets are incomplete, sunspot equilibria may exist.

Example. Since there is only one bond paying one unit of the numeraire in every state, the asset payoff vector is $A = (1, \dots, 1)'$. Hence, the span condition implies

$$p_s(x_s^i - \omega_s^i) = z^i \quad \text{for every state } s.$$

After splitting state 1 into two extrinsic states α, β , we have

$$p_\alpha(x_\alpha^i - \omega_1^i) = p_\beta(x_\beta^i - \omega_1^i) = z^i.$$

Thus, the same bond position must finance net trades in both extrinsic states. But this condition only restricts the value of net trades, not the commodity bundle itself. Therefore it does not force $x_\alpha^i = x_\beta^i$.

If equilibrium spot prices differ across the two extrinsic states, $p_\alpha \neq p_\beta$, then the same numeraire payoff can support different consumption bundles. Hence, with incomplete markets, it is possible that

$$x_\alpha \neq x_\beta$$

even though $\omega_\alpha = \omega_\beta = \omega_1$. This is the sunspot logic.

1.3.2 FWT under Asymmetric Information

Rationality of conjectures is critical for the First welfare theorem in the simple insurance economy.

Chapter 2

Abstract Exchange Economies

2.1 Pareto Efficient Allocations

Proposition 1 (Negishi Theorem). *(If) Let $\mathbb{U} = \{U \in \mathbb{R}^I \mid U \leq U(x) \text{ for some } x \in \mathbb{R}^{LI} \text{ such that } \sum_i x_i \leq \tilde{w}\}$ be a convex set. Let x be a Pareto efficient allocation. Then there exists $\alpha \in \mathbb{R}_+^I$ and $\alpha \neq 0$ such that x is the solution to the following problem:¹*

$$\begin{aligned} & \max_x \sum_i \alpha_i U_i(x_i) \\ & \text{subject to } \sum_i x_i \leq \tilde{w} \end{aligned}$$

(Only if) Given $\alpha \in \mathbb{R}_+^I$ and $\alpha \neq 0$, if x is the solution to the Negishi Problem, then x is Pareto Optimal.

Proof (If) Because $\mathbb{U}$ is convex and nonempty, and $U(x) \in \text{brdy}(\mathbb{U})$, by supporting hyperplane theorem, there exists α such that $\alpha U(x) \geq \alpha U'$ for all $U' \in \mathbb{U}$. Hence, $x = \arg\max_{x'} \sum_i \alpha_i U_i(x'_i)$. In addition, $\mathbb{U}$ is unbounded from below, we have $\alpha \in \mathbb{R}_+^I$.

(Only if) Suppose $\alpha \in \mathbb{R}_{++}^I$ and x solves the Negishi problem. Suppose, toward a contradiction, that x is not Pareto optimal. Then there exists a feasible allocation x' such that

$$U_i(x'_i) \geq U_i(x_i) \quad \forall i,$$

with strict inequality for some j . Since $\alpha_j > 0$, we have

$$\sum_i \alpha_i U_i(x'_i) > \sum_i \alpha_i U_i(x_i),$$

contradicting the optimality of x in the Negishi problem. Therefore, x is Pareto optimal.

¹In other words, there exists a Pareto weights that can justify the allocation.

■

2.2 Welfare Analysis

Proposition 2 (First Welfare Theorem). *In an abstract exchange economy characterized by $w \in \mathbb{R}_{++}^{S+1}$, any competitive equilibrium allocation $x \in \mathbb{R}_+^{S+1}$ is Pareto efficient.*

Proof: We prove by contradiction. Suppose there exists y that Pareto dominates x , then $y^i \succeq_P x^i \forall i$, and $y^j \succ_P x^j$ for some j . Then by the Walras Law and the Law of Revealed Preferences, we have

$$p_0(y_0^i - w_0^i) + \sum_s p_s(y_s^i - w_s^i) \geq p_0(x_0^i - w_0^i) + \sum_s p_s(x_s^i - w_s^i) = 0, \quad \text{and}$$

$$p_0(y_0^j - w_0^j) + \sum_s p_s(y_s^j - w_s^j) > p_0(x_0^j - w_0^j) + \sum_s p_s(x_s^j - w_s^j) = 0$$

Summing over i (i.e., summing the two equations above), we have

$$p_0 \sum_i (y_0^i - w_0^i) + \sum_s p_s \sum_i (y_s^i - w_s^i) > 0$$

Since p_0, p_s are strictly positive, there exists an s such that

$$\sum_i (y_s^i - w_s^i) > 0,$$

which implies that y is not feasible. ■

The converse of the First welfare theorem is the Second welfare theorem below. **Interpretation:** In words, the Second Welfare Theorem states that any Pareto efficient allocation can be decentralized as a competitive equilibrium via transfer (equivalently, redistribution of initial endowment, where the transferred endowment $\Delta w^i = \frac{T_i}{p}$) — See MWG Figure 15.b.13 for an illustration.

Proposition 3 (Second Welfare Theorem (The version with transfer)). *In an exchange economy with strictly monotonic, continuous, strictly quasiconcave U_i , and endowment $w \in \mathbb{R}_{++}^{LI}$: for any Pareto efficient allocation x , there exist balanced transfers t (such that $\sum_i t_i = 0$), and prices $p \in \mathbb{R}_{++}^L$ such that x is a competitive equilibrium of the economy $w + t$ at prices p .*

A Shorter Proof

- **Step 1: Define the transfers.** Let $t^i = x^i - w^i$. By feasibility, we have $\sum_i t^i = 0$. Now it remains to show that there exists a price system where x is a competitive equilibrium.
- **Step 2: Define the strict upper Contour set.** For x^i , define $B^i = \{\hat{x}^i : U^i(\hat{x}^i) > U^i(x^i)\}$. Define $B = \sum_i B^i$. Because utility is [strictly quasiconcave](#), B^i is convex and thus B is convex.

By Supporting hyperplane theorem, there exists $p \neq 0$ such that

$$p\hat{x} \geq p \sum_i x^i, \quad \forall \hat{x} \in B$$

- **Step 3: Prove that $p \in \mathbb{R}_{++}^L$.** Suppose there exists $p_l \leq 0$, by [strict monotonicity](#), we can increase the consumption of good l in some $\hat{x}^i \in B^i$ such that the resulting bundle $\hat{x} \in B$ but $p\hat{x} < p \sum_i x^i$. Contradiction. Hence, $p \in \mathbb{R}_{++}^L$.
- **Step 4: Prove x^i is the solution for all i .** Because U^i is [continuous](#) and $x^i \in \mathbb{R}_{++}^L$ ([interiority](#)), the cheaper-bundle argument upgrades the weak inequality to strict:

$$p\hat{x}^i > px^i, \quad \forall \hat{x}^i \in B^i$$

Hence, x^i is the maximizer for all i .

Q.E.D.

Proof Remark: The key of the proof is a separating hyperplane argument.

Let $x \in \mathbb{R}_{++}^{LI}$ be a Pareto Efficient allocation. Choose $w = x$. Define the strict upper contour set for agent i ,

$$B^i(x_i) := \{y_i \in \mathbb{R}_+^L : U(y_i) > U(x_i)\}$$

Define the aggregate strict upper contour set as

$$B(x) := \left\{ \sum_i y_i \in \mathbb{R}_+^L : y_i \in B^i(x_i) \forall i \right\}$$

Since $B^i(x_i)$ is convex, the aggregate upper contour set is also convex. In addition, by Supporting Hyperplane theorem, there exists a nonzero price vector p , such that

$$\inf_{a \in B(x)} pa \geq p \sum_i x_i$$

We will use this equation to establish that

- (1) $p \in \mathbb{R}_{++}^L$, and
- (2) under such price, no individual will trade.

Part 1. Suppose there exists some l such that $p_l < 0$, let

$$y_k^i = \begin{cases} x_k^i + \delta & \text{if } k = l \\ x_k^i & \text{if } k \neq l \end{cases}$$

Then, $py^i < px^i$ and $y^i \in B^i(x^i)$. Summing across agents, we have a contradiction to the supporting hyperplane condition above.

Now suppose there exists some l such that $p_l = 0$, let

$$y_k^i = \begin{cases} x_k^i + \delta & \text{if } k = l \\ x_k^i - \epsilon & \text{if } k \neq l \end{cases}$$

where δ is sufficiently large and ϵ is sufficiently small, so that the loss in $k \neq l$ can be compensated. Then, $py^i < px^i$ and $y^i \in B^i(x^i)$. Summing across agents, again, we have a contradiction to the supporting hyperplane condition above.

Part 2. Finally, we prove that under p found by the supporting hyperplane theorem, the equilibrium is autarky. Suppose there is a strictly profitable deviation from x^i for agent i , namely,

$$\exists \tilde{x}^i, \quad \text{s.t. } p\tilde{x}^i < px^i \quad \text{and} \quad \tilde{x}^i \in B^i(x^i)$$

Denote the payoff from the deviation by $\delta := px^i - p\tilde{x}^i > 0$. Pick $\tilde{x}^i + \sum_{j \neq i} y^j \in B(x)$, and apply the SHT, to obtain

$$p \left(\tilde{x}^i + \sum_{j \neq i} y^j \right) \geq p \sum_i x^i$$

It follows that

$$p \left(\sum_{j \neq i} y^j - x^j \right) \geq \delta > 0$$

However, consider $y^j = x^j + \epsilon$, we have $\lim py^j = \lim px^j$ as $\epsilon \rightarrow 0$, a contradiction. Hence, it cannot exist any profitable deviation for any agent i . ■

2.3 Existence Proof by Debreu's Map

Statement: A competitive equilibrium exists for any economy $\omega \in \mathbb{R}_{++}^{LI}$

Proof

Let $p \in \Delta^L$, $z : \text{Int}(\Delta^L) \rightarrow \mathbb{R}^L$, and $f : \Delta^L \rightrightarrows \Delta^L$ defined as follows. (This is Debreu's Map):

$$f(p) = \begin{cases} \text{argmax}_q qz(p) & \text{if } p \in \text{int}(\Delta^L) \\ \{q \in \Delta^L : pq = 0\} & \text{if } p \in \text{bdry}(\Delta^L) \end{cases}$$

This map implies that if there exists a fixed point, p^* then we have $z(p^*) = 0$ and $p^* \in \text{int}\Delta^L$. It remains to proof the existence of the fixed point. We can finish the proof by the Kakutani Fixed Point Theorem, because Δ^L is non-empty, convex, and compact. In addition, $f(p)$ is non-empty as

it maps p to either the interior or the boundary. $f(p)$ is also convex valued. Finally, by examining the following three cases, we can show that $f(p)$ is upper hemicontinuous.

1. $p^n \in \text{Int}(\Delta^L) \rightarrow p \in \text{Int}(\Delta^L)$
2. $p^n \in \text{bdry}(\Delta^L) \rightarrow p \in \text{bdry}(\Delta^L)$
3. $p^n \in \text{Int}(\Delta^L) \rightarrow p \in \text{bdry}(\Delta^L)$

Importantly, in case 3 above, we need to use the boundary condition of aggregate excess demand. Namely,

$$\text{If } p^n \rightarrow p \text{ with some } p_l = 0, \text{ then } \max\{z_1(p^n), \dots, z_L(p^n)\} \rightarrow \infty$$

By the Kakutani FPT, there exists $p^* \in f(p^*)$. *Q.E.D.*

2.4 Sufficient conditions for the uniqueness of competitive equilibrium

Existence of a competitive equilibrium holds under general conditions, but uniqueness requires strong restrictions. Proposition 6 of the notes provides three sufficient conditions:

1. **Pareto-efficient endowments.** If ω is Pareto efficient, the unique equilibrium is autarchic ($x^i = \omega^i$ for all i), since there are no gains from trade and strict convexity of preferences pins down the allocation on the budget line.
2. **Identical, homothetic preferences.** If all agents share the same homothetic U^i , then $x^i(p, y^i) = y^i x(p, 1)$ aggregates linearly, so the economy admits a representative agent owning $\sum_i \omega^i$, and equilibrium prices depend only on aggregate endowments.
3. **Gross substitutes.** If aggregate excess demand satisfies

$$p'_l > p_l \text{ and } p'_j = p_j \ \forall j \neq l \implies z_j(p') > z_j(p),$$

then raising prices coordinate-by-coordinate from one candidate equilibrium p to another p' contradicts $z(p) = z(p') = 0$.

Caveat. These conditions are sufficient but not necessary; by the Sonnenschein–Mantel–Debreu theorem, aggregate excess demand can take essentially any shape, so multiple equilibria are generic absent such restrictions.

2.5 Sonnenschein-Debreu-Mantel Theorem (Anything-goes)

The following famous theorem is known as the “anything-goes” theorem and Indeterminacy Theorem, in the sense that, one can take “any” arbitrary aggregate excess demand function (z), and construct an economy with rational individuals that matches it. This proves that individual rationality imposes no restrictions on the aggregate market behavior. This undertermineness at the macro level outcomes leads to a famous quote by Tom Sargent:

“The real problem in economics is not that our models impose too many restrictions on the data, but far too few”.

Jesús Fernández-Villaverde says: SMD teaches us that meaningful economic predictions require disciplined structure beyond individual rationality.

Proposition 4 (Sonnenschein-Debreu-Mantel Theorem). *Let Δ_ϵ^{L-1} be a trimmed simplex, with $\epsilon > 0$. Given any continuous function $z : \Delta_\epsilon^{L-1}$ satisfying continuity and Walras law, then there is an economy with $U^i : \mathbb{R}_+^L \rightarrow \mathbb{R}$, $\omega \in \mathbb{R}_{++}^{LI}$, $I \geq L$, whose aggregate excess demand function, restricted to Δ_ϵ^{L-1} , equals z .*

2.6 Consumption Externalities and Pigouvian Taxes

In the presence of consumption externalities, an individual’s utility depends on both her and others’ consumptions. As a result, the centralized allocation (i.e., Negishi’s solution) deviates from the decentralized one. To induce the decentralized allocation to be Pareto Efficient, one can introduce the so-called Pigouvian taxes, so that the taxes make individuals behave *as if* they care about others. The following derives the tax formula.

For simplicity, we assume that the goods 1 does not have consumption exteranlity. First, the Negishi problem is give by²

$$\begin{aligned} & \max_{x \in \mathbb{R}^{LI}} \sum_i \alpha^i U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) \\ & \text{subject to: } \sum_i x_l^i \leq w_l, \quad \forall l \end{aligned}$$

Note that there are L constraints, so we introduce L Lagrangean Multipliers in the equation below

$$\mathcal{L} = \sum_i \alpha^i U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) + \sum_l \lambda_l \left(w_l - \sum_i x_l^i \right)$$

²We use superscripts to denote individuals, and subscripts to index goods.

The first-order conditions are given by

$$\begin{aligned} [x_1^i] \quad & \alpha^i \frac{\partial U^i}{\partial x_1^i} = \lambda_1 \\ [x_l^i] \quad & \sum_j \alpha^j \frac{\partial U^j}{\partial x_l^i} = \lambda_l \quad \text{for } l \neq 1 \end{aligned}$$

The decentralized consumer's problem is given by

$$\begin{aligned} & \max_{x^i \in \mathbb{R}^I} U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) \\ \text{subject to: } & \sum_l p_l (1 + \tau_l^i) x^i \leq \sum_l p_l w_l^i, \quad \forall l \end{aligned}$$

The Lagrangean is given by

$$\mathcal{L}^i = U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) + \mu^i \sum_l [p_l w_l^i - p_l (1 + \tau_l^i) x^i]$$

The first-order conditions are given by

$$\begin{aligned} [x_1^i] \quad & \frac{\partial U^i}{\partial x_1^i} = \mu^i p_1 = \mu^i \quad \text{normalization using } p_1 = 1 \\ [x_l^i] \quad & \frac{\partial U^i}{\partial x_l^i} = \mu^i p_l (1 + \tau_l^i) \quad \text{for } l \neq 1 \end{aligned}$$

Combining the these two conditions, we have

$$\frac{\partial U^i}{\partial x_l^i} = \frac{\partial U^i}{\partial x_1^i} p_l (1 + \tau_l^i)$$

Note that we can decompose the second FOC of the Negishi problem as

$$\frac{\partial U^i}{\partial x_l^i} + \sum_{j \neq i} \frac{\alpha^j}{\alpha^i} \frac{\partial U^j}{\partial x_l^i} = \frac{\lambda_l}{\alpha^i} \quad \text{for } l \neq 1$$

Therefore, we can equate the equation above with the decentralized condition and compute the tax formula:

$$\underbrace{\frac{\lambda_l}{\alpha^i} - \sum_{j \neq i} \frac{\alpha^j}{\alpha^i} \frac{\partial U^j}{\partial x_l^i}}_{\text{Negishi's}} = \underbrace{\frac{\partial U^i}{\partial x_1^i} p_l (1 + \tau_l^i)}_{\text{Decentralized consumer's}}$$

It follows that

$$\tau_l^i = \frac{\lambda_l}{p_l \alpha_i \frac{\partial U^i}{\partial x_1^i}} - 1 - \sum_{j \neq i} \frac{\alpha^j}{p_l \alpha^i} \frac{\frac{\partial U^j}{\partial x_l^i}}{\frac{\partial U^i}{\partial x_1^i}}$$

Now assuming $\alpha^i = \frac{1}{\mu^i} = \frac{1}{\frac{\partial U^i}{\partial x_1^i}}$, and $p_l = \lambda_l$, then we have

$$\tau_l^i = -\frac{1}{p_l} \sum_{j \neq i} \frac{\frac{\partial U^j}{\partial x_l^i}}{\frac{\partial U^j}{\partial x_1^j}}.$$

2.7 Problems

2.7.1 Problem 1.1

Problem. Show that, in an economy with consumption externalities, the proof of the First Welfare theorem breaks down. Furthermore, in this economy, compare the first order conditions of the Negishi and those of the Consumer pb and argue that — in general — they cannot be all satisfied at an allocation $x \in \mathbb{R}_+^{LI}$. Furthermore,

1. Consider taxing consumption x^i at the rate τ^i , for each $i \in I$. Let $\tau = [\tau_l^i]_{i \in I}$. The budget constraint of agent $i \in I$ can be written as

$$p \square (1 + \tau^i) x^i \leq p \omega^i + T^i,$$

where T^i is the lump-sum re-batement of tax proceeds to agent i . Show that the tax rate

$$\tau_l^i = -\frac{1}{p_l} \sum_{j \neq i} \frac{\partial U^j(x^j, x^{-j}) / \partial x_l^i}{\partial U^j(x^j, x^{-j}) / \partial x_1^j}, \quad \forall i \in I, l = 2, \dots, L \quad (\text{Pigouvian tax})$$

induces a competitive equilibrium allocation which is Pareto Efficient.

2. Consider requiring that any consumption allocation x^i be associated to the acquisition of consumption rights r^i — at price p^r — to be actually consumed. The market clearing conditions $x^i = r^i$ should then also be satisfied at a competitive equilibrium. The budget constraint of agent $i \in I$ can be written as

$$p x^i + p^r r^i \leq p \omega^i.$$

Show that at prices

$$p_l^{r,i} = \sum_{j \neq i} \frac{\partial U^j(x^j, x^{-j}) / \partial x_l^i}{\partial U^j(x^j, x^{-j}) / \partial x_1^j}, \quad \forall i \in I, l = 2, \dots, L \quad (\text{Lindahl prices})$$

the competitive equilibrium allocation is Pareto Efficient.

Solution. (a) The proof breaks down because we cannot conclude that $y \succ^P x$ implies $\sum_{i \in I} p y^i > \sum_{i \in I} p x^i$. The FOCs of the Consumer problem are (for all $i \in I$ and $\ell \in L$):

$$\frac{\partial U^i}{\partial x_\ell^i} = \lambda^i p_\ell.$$

The FOCs of the Negishi problem are (for all $i \in I$ and $\ell \in L$):

$$\alpha^i \frac{\partial U^i}{\partial x_1^i} = \mu_1, \quad \sum_{j \in I} \alpha^j \frac{\partial U^j}{\partial x_\ell^j} = \mu_\ell.$$

In general, they do not coincide.

(b) The FOCs of the Consumer problem become (for all $i \in I$ and $\ell \in L$):

$$\begin{aligned} \frac{\partial U^i}{\partial x_1^i} &= \lambda^i, \\ \frac{\partial U^i}{\partial x_\ell^i} &= \lambda^i p_\ell (1 + \tau_\ell^i) = \lambda^i p_\ell - \lambda^i \sum_{j \neq i} \frac{\partial U^j / \partial x_\ell^i}{\partial U^j / \partial x_1^j} = \lambda^i \left(p_\ell - \sum_{j \neq i} \frac{\partial U^j / \partial x_\ell^i}{\lambda^j} \right). \end{aligned}$$

Therefore:

$$\frac{\partial U^i}{\partial x_1^i} = \lambda^i, \quad \sum_{j \in I} \frac{1}{\lambda^j} \frac{\partial U^j}{\partial x_\ell^j} = p_\ell.$$

These are the same FOCs as the Negishi problem's, up to re-labeling ($\alpha^j = \lambda^j$, $\mu_\ell = p_\ell$).

(c) Clearly, it is sub-optimal to waste money on $r^i > x^i$. Therefore, the market clearing condition $x^i = r^i$ is redundant as it is implied by monotonicity and individual feasibility. Let τ^i denote the Pigouvian tax. The budget constraint of agent $i \in I$ can be re-written as:

$$p\omega^i \geq p x^i + p^r r^i = p x^i + p^{r,i} x^i = p \square \left(1 + \frac{1}{p} \square p^{r,i} \right) x^i = p \square (1 + \tau^i) x^i.$$

Therefore, Lindahl prices have the same effect as the Pigouvian tax.

Chapter 3

Abstract Production Economies

3.0.1 Competitive equilibrium

Definition. For an economy with endowment $(\omega, \delta) \in \mathbb{R}_{++}^{LI}, \Delta^{IH}$, a competitive equilibrium is $(x, y, p) \in \mathbb{R}_+^{LI} \times Y \times \mathbb{R}_{++}^L$, such that

1. Given $p \in \mathbb{R}_{++}^L$, each agent i solves

$$\begin{aligned} & \max_{x^i} U^i(x^i) \\ \text{subject to } & px^i \leq pw^i + \sum_h \delta^{ih} \pi^h(p) \end{aligned}$$

2. Given p , each firm h solves

$$\max_{y^h} py^h$$

3. Goods market clears:

$$\sum_i x^i \leq \sum_i w^i + \sum_h y^h$$

4. consistency. Namely, profit is given by

$$\pi^h(p) = py^h, \forall h \in H$$

3.1 Problems

3.1.1 Problem 2.3

Problem. Prove existence of a Romer competitive equilibrium; that is, show that the standard proof applies once it is handled with care.

Solution. For every fixed $K = (K^h)_{h \in H}$, we can follow the standard proof to show that a competitive equilibrium exists (however it may not satisfy consistency, so it may not be a Romer CE). Denote by $(x(p, w), y(p, w, K), k(p, w, K))$ the equilibrium induced by (p, w, K) . Finding a Romer CE means finding a fixed point of $k_{p,w}(K)$. One can check that $k_{p,w}$ is well-behaved so that Kakutani's FP Theorem applies.

3.1.2 Problem 2.4

Problem. Drawing by analogy from Problem 1.2 on public goods show that, for a Romer competitive equilibrium, the proof of the First Welfare theorem breaks down. Furthermore, in this economy, compare the first order conditions of the PE pb and those of the Firm pb and argue that — in general — they cannot be all satisfied at a production plan $(y, k) \in \mathbb{R}_+^H \times \mathbb{R}_+^{(L-1)H}$. Can you construct a constrained notion of Pareto efficiency for which the First Welfare Theorem holds in this economy? [Hint: Write the corresponding Negishi problem]

Solution. The proof breaks down because we cannot conclude from $(x, y, k) \succ^P (x_*, y_*, k_*)$ that $\sum_{h \in H} p y^h > \sum_{h \in H} p y_*^h$. The FOCs (wrt k) of the Firm's problem are (for all $h \in H$ and $\ell \neq 1$):

$$pA(K^h) \frac{\partial f^h}{\partial k_\ell^h} = 0.$$

The FOCs (wrt k) of the Negishi problem are (for all $h \in H$ and $\ell \neq 1$):

$$pA(k^h) \frac{\partial f^h}{\partial k_\ell^h} + p \frac{\partial A}{\partial k_\ell^h} f^h(k^h) = 0.$$

This happens because the planner internalizes the consistency requirement. A constrained notion of efficiency would require the planner to maximize welfare taking K^h as fixed:

$$\max_{x, y, k} \sum_{i \in I} \alpha^i U(x^i) \quad \text{s.t.} \quad \begin{cases} \sum_{i \in I} x_\ell^i + \sum_{h \in H} k_\ell^h \leq \sum_{i \in I} w_\ell^i \\ \sum_{i \in I} x_1^i \leq \sum_{i \in I} w_1^i + \sum_{h \in H} y^h \\ y^h = A(K^h) f^h(k^h) \end{cases}$$

Chapter 4

Financial Market Economies

4.1 Arrow-Debreu Equilibrium

Definition 4.1.1. An Arrow-Debreu equilibrium is defined by an allocation $x_0, \{x_s\}$, such that

1. given the goods prices ϕ_0, ϕ_s , the allocation solves the consumer's problem below:

$$\begin{aligned} \max_{x_0, \{x_s\}} \quad & u_0(x_0) + \sum_s \text{Prob}_s u_s(x_s) \\ \text{s.t.} \quad & \phi_0(x_0 - w_0^i) + \sum_s \phi_s(x_s - w_s^i) = 0 \end{aligned}$$

2. Goods market clears:

$$\sum_i x^i \leq \sum_i w^i$$

4.2 No-Arbitrage

There are multiple definitions of the No Arbitrage condition. Essentially, they all require zero profit on the spot market at time 0, and the future financial market no matter what the future state will be. Formally, the first definition is given by

Definition I (No Arbitrage): Let $A \in \mathbb{R}^{S \times J}$ be a future return matrix. Let $z \in \mathbb{R}^J$ be a portfolio. $q^T \in \mathbb{R}^J$ be a price vector of the portfolio.^a For $W = \begin{bmatrix} -q \\ A \end{bmatrix}$, there is no $z \in \mathbb{R}^J$ such that $Wz > 0$.

- **Remark:** Intuitively, $Wz \in \mathbb{R}^{S+1}$ denotes the profit of buying/selling assets on the spot market and all the future financial markets.

^aIn this note, only q is a row vector, while other vectors are columns.

Example: Consider $W = \begin{bmatrix} -3 & -4 \\ 5 & 0 \\ 2 & 2 \\ 3 & 1 \end{bmatrix}$. Then take $z = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (namely, buying 1 unit of asset 1 and

selling 1 unit of asset 2), we have $Wz = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 2 \end{bmatrix}$, which means the agent can make profit 1 at time 0 (on the spot market), and 5, 0, 2 at time 1 if $s = 1, 2, 3$, respectively.

The No Arbitrage condition above is (mathematically) equivalent to the following definition

Definition II (No Arbitrage):

$$\text{span}(W) \cap \mathbb{R}_+^{S+1} = \{\mathbf{0}\},$$

where $\text{span}(W) = \{Wz, \forall z \in \mathbb{R}^J\}$ denotes the profit of all possible portfolio.

With these two definitions in mind, we now introduce the No-Arbitrage theorem

Defintion (No-Arbitrage Theorem)

$$\text{span}(W) \cap \mathbb{R}_+^{S+1} = \{\mathbf{0}\} \Leftrightarrow \exists \hat{\pi} \in \mathbb{R}_{++}^{S+1} \text{ such that } \hat{\pi}\tau = 0, \forall \tau \in \text{span}(W)$$

Remarks:

- We require $\hat{\pi} \in \mathbb{R}_{++}^{S+1}$ (strictly positive) due to the need of normalization (see below).

Intuitively, $\hat{\pi}$ serves as a machinery that summarizes the agent's profit at time 0 and time 1,

taking probabilities and discount factors into considerations:

$$\hat{\pi}_0[Wz]_0 + \sum_s \hat{\pi}_s[Wz]_s = 0, \quad \forall z \in \mathbb{R}^J$$

Since $[Wz]_0 = -\sum_j q_j z_j$, after the normalization $\pi_s = \frac{\hat{\pi}_s}{\hat{\pi}_0}$, we have

$$-\sum_j q_j z_j + \sum_s \pi_s \sum_j a_{sj} z_j = 0,$$

where π_s denotes the price of one unit of consumption good delivered in state s at time 1. In matrix form, we have $(-q + \pi^T A)z = 0 \Rightarrow q = \pi^T A$.

Defining $\pi_s = m_s \text{Prob}_s$, we have the **Fundamental Equation of Asset Pricing**:

$$q = \mathbb{E}(mA). \tag{4.2.1}$$

Definition (No-Arbitrage Theorem).

$$\text{span}(W) \cap \mathbb{R}_+^{S+1} = \{\mathbf{0}\} \iff \exists \hat{\pi} \in \mathbb{R}_{++}^{S+1} \text{ such that } \hat{\pi} \cdot \tau = 0, \quad \forall \tau \in \text{span}(W),$$

where $W = \begin{bmatrix} -q \\ A \end{bmatrix}$.

Remarks:

- Alternatively, we can state the theorem as follows: If an asset price vector q admits no arbitrage, then there exists a state price vector $\pi \in \mathbb{R}_{++}^S$ such that

$$q = \pi A.$$

- **Equivalence with $q = \pi A$.** Writing $\hat{\pi} = (\hat{\pi}_0, \pi) \in \mathbb{R}_{++} \times \mathbb{R}_{++}^S$, the orthogonality condition $\hat{\pi} \cdot Wz = 0$ for all $z \in \mathbb{R}^J$ gives

$$(-\hat{\pi}_0 q + \pi A)z = 0 \quad \forall z \in \mathbb{R}^J \iff \hat{\pi}_0 q = \pi A.$$

Normalizing $\hat{\pi}_0 = 1$ recovers the familiar form

$$q = \pi A,$$

with $\pi \in \mathbb{R}_{++}^S$.

- Although the alternative version is more readable, the W -version helps the proof. See below.

Proof of the No-Arbitrage Theorem

($\Rightarrow$) We prove this direction in two steps:

1. “Strict positivity”. First, we define a simplex in the space of $\mathbb{R}^{S+1}$:

$$\Delta = \{\tau \in \mathbb{R}^{S+1} : \sum_{s=0}^S \tau_s = 1\}$$

By the definition of No-Arbitrage condition (i.e., the LHS of the theorem), we have

$$\text{span}(W) \cap \Delta = \emptyset$$

Now, invoking the **Strong Separating Hyperplane Theorem** (this is the essence of

the proof), then there exists $\hat{\pi}$, such that

$$\sup_{\tau \in \text{span}(W)} \hat{\pi}\tau < \inf_{\tau \in \Delta} \hat{\pi}\tau \quad (4.2.2)$$

To prove $\hat{\pi} \in \mathbb{R}_{++}^{S+1}$, suppose there exists a $\hat{\pi}$ whose s^{th} component, $\pi_s \leq 0$, then since the base vector ($e_s \in \Delta$), we have

$$\inf_{\tau \in \Delta} \hat{\pi}\tau \leq e_s\tau \leq 0$$

However, $0 \in \text{span}(W)$ (in words, no buying and selling in the asset market), which implies that

$$\sup_{\tau \in \text{span}(W)} \hat{\pi}\tau \geq \hat{\pi}0 = 0$$

Hence, we have the $0 \leq LHS < RHS \leq 0$ in equation (??), a contradiction.

2. **“Orthogonality”**.. Suppose there exists a $\tau \in \text{span}(W)$ such that $\hat{\pi}\tau \neq 0$. Then, we can find an α such that $\hat{\pi}\alpha\tau$ is arbitrarily large. (In words, agent can make infinite profit if there is an opportunity for arbitrage). However, the RHS of the theorem is bounded above (because Δ is a compact set). Then we have a contradiction. Hence,

$$\hat{\pi}\tau = 0 \forall \tau \in \text{span}(W)$$

($\Leftarrow$) Now suppose there exists an arbitrage opportunity $\tau^* \in \text{span}(W)$ such that $\tau^* \in \mathbb{R}^{S+1} \setminus \{0\}$. Since $\hat{\pi} \in \mathbb{R}_{++}^{S+1}$, we have $\hat{\pi}\tau^* > 0$. However, this contradicts to the assumption that $\hat{\pi}\text{span}(W) = 0$ as $\tau^* \in \text{span}(W)$. ■

Proposition. (The degree of freedom of stochastic discounts) Given $A \in \mathbb{R}^{S \times J}$ of rank J and $q \in \mathbb{R}^J$, the set of viable stochastic discounts is an $\mathbb{R}^{S-J}$ strictly positive subspace of $\mathbb{R}^J$, namely,

$$R(q; A) = \{\pi \in \mathbb{R}_{++}^S : q = \pi^T A\} = \mathbb{R}_{++}^{S-J}$$

In words, once you find a viable π , there are only $S - J$ elements in that π you can vary.

4.3 Arrow Theorem

4.3.1 Budget constraints equivalence

Proposition 5. *The following two budget constraints are equivalent under No-Arbitrage:*

1. $p_0(x_0^i - w_0^i) + qz = 0$ and $p_s(x_s^i - w_s^i) = a_s z$

2. $p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s(x_s^i - w_s^i) = 0$ and,

$$\begin{bmatrix} \vdots \\ p_s(x_s^i - w_s^i) \\ \vdots \end{bmatrix} \in \text{span}(A)$$

Remarks: When the asset market is complete $\text{span}(A) = \mathbb{R}^J = \mathbb{R}^S$.

Proof Suppose q satisfies the No-Arbitrage condition, we have the following asset pricing equation

$$q = \pi A, \quad \text{with } q \in \mathbb{R}^J, \pi \in \mathbb{R}^S, A \in \mathbb{R}^{S \times J}$$

Now the first equation of the first set of constraints becomes

$$p_0(x_0^i - w_0^i) + qz = p_0(x_0^i - w_0^i) + \pi Az = 0,$$

which is equivalent to

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s a_s z = 0$$

Using the the second equation of the first set of constraints to replace $a_s z$, we have

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s(x_s^i - w_s^i) = 0$$

To complete the proof, notice that the second equation of the second set of constraints is simply the matrix for of

$$p_s(x_s^i - w_s^i) = a_s z$$

$\Rightarrow$

$$\begin{bmatrix} \vdots \\ p_s(x_s^i - w_s^i) \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ a_s \\ \vdots \end{bmatrix} z$$

which belongs to the span of A . ■

When $L = 1$, we can use the above statement about “constraint equivalence” to show that the constrained set of consumption $B^i \perp p$, and thus we have constrained P.E. Namely,

$$\left(B^i := \begin{cases} x_0^i - w_0^i + \sum_{s=1}^S \pi_s (x_s^i - w_s^i) = 0 \\ [(x_s^i - w_s^i)]_{\forall s} \in \langle A \rangle \end{cases} \right) \perp q$$

Notice that we use $q = \pi A$ in the derivation of the first equation of B^i . However, in the presence of transaction costs, $q \leq \pi A \leq q + \gamma$, the tricks we used in the proof of Proposition ?? break down. Hence, $B^i(q)$ is a function of q when $\gamma > 0$.

4.3.2 Arrow’s Theorem

Theorem 1 (Arrow’s Theorem). *Under the No-Arbitrage condition, any Arrow-Debreu equilibrium $(x_0^*, \{x_s^*\}, \{p_0^*, p_s^*\})$ can be supported as a Financial Market Equilibrium (FME) $(x_0^*, \{x_s^*\}, z^*, \{p_0^*, q^*\})$, where z^* is the portfolio that satisfies the budget constraints in the FME, and q^* satisfies the No-Arbitrage condition.*

Proof Since goods market clearing implies financial market clearing, the market clearing conditions in the FME is the same as the one in Arrow-Debreu equilibrium. Now, we only need to discuss the equivalence of the budget constraints between these two economies.

Since the market is complete, the span condition in the FME is not binding. Then the budget constraints in the FME reduce to

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s (x_s^i - w_s^i) = 0$$

Now using $p_0 = \phi_0$ and $p_s = \pi_s \phi_s$, we can immediately see the equivalence between the AD equilibrium and the FME. ■

4.4 Non-convexity

Proposition 6. *In economies with non-convex commodity space, an equilibrium may fail to exist.*

Proof (Sketch): Consider an economy with non-convex commodity space, for example, when the quantities of some goods are discrete (namely, indivisible goods). Then, the aggregate excess demand function may not be continuous. To tackle this issue, one can consider the convex hull of the aggregate excess demand function. By Kakutani’s FPT, we have $0 \in \text{conv}(Z(\pi, p))$. If $0 \in Z(\pi, p)$, then we are done. Otherwise, we can apply the Caratheodory’s theorem to find $S + 1 + 1$ vector such that $\sum \alpha_k z_k = 0$ with $z_k \in Z(\pi, p)$, $\alpha_k \geq 0$, $\sum \alpha_k = 1$. Since agent i is indifferent among $\{z_{ik}\}$ across k , we can assign α_k fraction of type i agents to choose z_{ik} . As a result, the aggregate excess

demand is zero. ■

4.5 Cass Trick

Question: What is the Cass trick and how is it useful in the existence proof for (single commodity, two-periods) financial market economies with incomplete markets?

Answer: The definition of the Cass trick. Consider an economy where the span condition is imposed on all consumers except $i = 1$. Then, the Cass trick argues that the Financial market equilibrium (FME) is the same as the “modified” FME. We use the conclusion of the Cass trick to establish the boundary condition of the FME. Specifically, for $i = 1$, her excess demand goes to infinity as price goes to zero. Using the budget constraints for other individuals, we can see that the excess demand is bounded from below. Hence, using the same proof as in the existence proof of the AD-equilibrium, we can establish the existence.

4.6 Cass Trick in a 3-period FME

Want: The existence proof of a competitive equilibrium in a 3-period Financial Market Equilibrium.

Sketch: First, we can extend the Lemma of constraint equilvance to a 3-period setup, which includes $1 + S + S^2$ budget constraints (i.e., normalizations). The S^2 additional equations come from the **spanning condition** at period $t = 1$. This extended theorem reduces the price space to $(\pi, p) \in \mathbb{R}_{++}^{S+S^2} \times \mathbb{R}_{++}^{L(1+S+S^2)}$, where π is the state price and p is the spot price.

In addition, the market clearing conditions are

$$\sum_i x^i(\pi, p) - w^i = 0$$

The total number of market clearing conditions are $L(1 + S + S^2) - 1$ because there are L goods, 1 spot market today, S tomorrow and S^2 the day after tomorrow. -1 because of the Walras Law.

Then we can count the total number of unknowns and equations:

$$\underbrace{S + S^2 + L(1 + S + S^2)}_{\text{raw unknowns}} - \underbrace{1 + S + S^2}_{\text{normalization}} = \underbrace{L(1 + S + S^2) - 1}_{\text{\# of mkt clearing cond.}}$$

The role played by the Cass trick is to eliminate the spanning conditions in $t = 1, 2$ for agent $i = 1$. So that the boundary condition holds and the equilibrium exists under this modified equilibrium. The last piece of the proof is just to show that the equivalence between the modified and the original equilibria.

4.7 Aggregation

As many economic analysis are conducted under the representative agent framework, such as the canonical consumption-based CAPM model, it is important to understand under what conditions we can aggregate agents' behavior.

Proposition 7. *If utility functions are identical and homothetic, we have the equilibrium prices **independent** of the distribution of endowments across agents. (In other words, p is a function of only aggregate endowment $\sum_i w^i$.)*

Proof of proposition ?? (Sketch): Define demand function $x_s^i(p_s, y_s^i)$. By Walras' law, we have $p_s x_s^i(p_s, y_s^i) = y_s^i$, and $\sum_i y_s^i = p_s \sum_i x_s^i = p_s \sum_i w_s^i$. Then, we have

$$\begin{aligned}
 \sum_i p_s x_s^i(p_s, y_s^i) &= \sum_i y_s^i \\
 \Rightarrow \sum_i p_s x_s(p_s, y_s^i) &= p_s \sum_i w_s^i \\
 \Rightarrow p_s \sum_i y_s^i x_s(p_s, 1) &= p_s \sum_i w_s^i \\
 \Rightarrow \sum_i y_s^i x_s(p_s, 1) &= \sum_i w_s^i \\
 \Rightarrow x_s(p_s, \sum_i y_s^i) &= \sum_i w_s^i \\
 \Rightarrow x_s(p_s, p_s \sum_i w_s^i) &= \sum_i w_s^i
 \end{aligned}$$

Therefore, p_s is uniquely determined by the equation above. ■

4.8 Problems

4.8.1 Problem 4.1

Problem. Consider an alternative definition of Constrained Pareto optimality, due to Grossman (1970), in which the spanning constraint is evaluated at prices $p_s \in \mathbb{R}^L$ (for any $s \in S$) representing competitive equilibrium prices of the Financial market economy $\omega \in \mathbb{R}_{++}^{L(S+1)I}$ with Incomplete markets ($\text{rank}(A) = J < S$). Show that the allocation $x \in \mathbb{R}_+^{L(S+1)I}$ pertaining to a Financial market equilibrium (x, π, p) is in fact Constrained Pareto efficient according to this definition.

Solution. Let (x, π, p) be a FM equilibrium. Suppose that there exists some feasible allocation y that Pareto dominates x and satisfies the market constraint. Since y satisfies the market constraint

but is not an equilibrium, it must be ruled out by the WLRP. Therefore,

$$\sum_{i \in I} \left[p_0(y_0^i - x_0^i) + \sum_{s \in S} p_s(y_s^i - x_s^i) \right] > 0.$$

But this contradicts $\sum_{i \in I} x^i = \sum_{i \in I} \omega^i = \sum_{i \in I} y^i$.

4.8.2 Problem 4.2

Problem. Consider a Complete market economy — with $\text{rank}(A) = S$ — whose feasible set of asset portfolios is restricted by borrowing limits:

$$z^i \geq -b^i, \quad \text{for any } i = 1, \dots, I.$$

Construct the implied constrained set of consumption allocations in this economy and show that it has the form $B^i(p_1, \dots, p_S)$.

Solution. We can write the set of feasible bundles as:

$$B^i(p) = \{x^i : \exists z^i \geq -b^i, p_0(x_0^i - \omega_0^i) + qz^i \leq 0 \wedge \forall s \in S, p_s(x_s^i - \omega_s^i) \leq a_s z^i\}.$$

4.8.3 Problem 4.3

Problem. Consider a Financial Market economy with one single commodity ($L = 1$) which lasts $t = 3$ periods. Agents trade the same J assets with payoff A at time $t = 0$ and time $t = 1$. Assume markets are incomplete, that is, $\text{rank}(A) < S$. Define a Financial market equilibrium for this economy as well as Constrained Pareto optimality. Construct the implied constrained set of consumption allocations in this economy and show how it depends on prices.

Solution. A FM equilibrium is a tuple $(x, z, q) \in \mathbb{R}^{1+S+S^2} \times \mathbb{R}^{J(1+S)} \times \mathbb{R}^{J(1+S)}$ that clears markets and:

$$\begin{aligned} \max_{x^i, z^i} & u^i(x_0^i) + \sum_{s \in S} \mathbb{P}(s) \left[u^i(x_{1s}^i) + \sum_{s' \in S} \mathbb{P}(s' | s) u^i(x_{2ss'}^i) \right] \\ \text{s.t.} & \begin{cases} x_0^i - \omega_0^i + q_0 z_0^i \leq 0 \\ x_{1s}^i - \omega_{1s}^i + q_{1s} z_{1s}^i \leq a_s z_0^i \\ x_{2ss'}^i - \omega_{2ss'}^i \leq a_s z_{1s}^i \end{cases} \end{aligned}$$

Re-write the set of constraints using the no-arbitrage condition:

$$\begin{cases} x_0 - \omega_0 + \sum_{s \in S} \pi_s^0 (x_{1s} - \omega_{1s}) + \sum_{s \in S} \sum_{s' \in S} \pi_{s'}^s (x_{2ss'} - \omega_{2ss'}) = 0 \\ (x_{2ss'} - \omega_{2ss'})_{s' \in S} \in \langle A \rangle \\ \exists z_1 : (x_s - \omega_s + \pi^s A z_{1s})_{s \in S} \in \langle A \rangle \end{cases}$$

So market incompleteness is characterized by the restriction:

$$B(\pi) = \{x^i : \exists z_1, (x_s - \omega_s + \pi^s A z_{1s})_{s \in S} \in \langle A \rangle \ni (x_{2ss'} - \omega_{2ss'})_{s' \in S}\}.$$

The Planner's problem is:

$$\begin{aligned} \max_{(x, \theta) \in \mathbb{R}^{I(1+S+S^2)} \times \mathbb{R}^{IJ(1+S)}} \sum_{i \in I} \alpha^i & \left[u^i(x_0^i) + \sum_{s \in S} \mathbb{P}(s) \left[u^i(x_{1s}^i) + \sum_{s' \in S} \mathbb{P}(s' | s) u^i(x_{2ss'}^i) \right] \right] \\ \text{s.t.} \quad & \begin{cases} \sum_{i \in I} x^i = \sum_{i \in I} \omega^i \\ x^i \in x^i(q, \omega, \theta) \\ q \in q(\omega, \theta) \end{cases} \end{aligned}$$

4.8.4 Problem 4.4

Problem. Consider a Complete market economy with $\text{rank}(A) = S$. Assume that, for an exogenous vector $\gamma \in \mathbb{R}_{++}^J$: the buying price of asset j is $q_j + \gamma_j$ while the selling price of asset j is q_j , for any $j = 1, \dots, J$. Let z_{j+}^i, z_{j-}^i denote, respectively, the buying and selling position of agent i on asset j . Suppose that trade occurs in equilibrium, that is $z_{j+}^i > 0$ for some $i \in I$ and some $j \in J$. Construct the implied constrained set of consumption allocations in this economy and show how it depends on prices.

Bonus questions: Can at a Financial market equilibrium be the case that both $z_{j+}^i, z_{j-}^i > 0$? Are Financial market equilibria Pareto efficient?

Solution. The set of feasible bundles can be written as:

$$B^i(p) = \left\{ x^i : \exists z_+^i, z_-^i, \begin{cases} p_0(x_0^i - \omega_0^i) + (q + \gamma)z_+^i - qz_-^i \leq 0 \\ p_s(x_s^i - \omega_s^i) \leq a_s(z_+^i - z_-^i) \end{cases} \right\}.$$

In equilibrium it is impossible that $z_+^i, z_-^i > 0$, as it would be strictly preferable to either sell or buy $|z_+^i - z_-^i|$, depending on the sign of the difference. Equilibria are not Pareto efficient because the market constraint depends on prices.

4.8.5 Problem 4.5

Problem. Consider an economy with I agents, S states, $L > 1$ commodities, and a full set of Arrow securities: $A = I_S$. Agents can default in each state $s \in S$, after the state is realized. If they default they consume only a fraction $\alpha \in (0, 1)$ of their endowment. The remaining fraction, $1 - \alpha$, is first pooled across all defaulting agents and then re-distributed pro-rata to their creditors.

Write down a constraint on agents' portfolios of Arrow securities which guarantees that they will not default. Construct the implied constrained set of consumption allocations guaranteeing no default and show how it depends on prices. Will agents ever default at a Constrained Pareto efficient allocation? Is there anything fundamental about Arrow securities — or about complete markets — which drives your answers?

Solution. The constraint $z_s^i \geq -(1 - \alpha)\omega_s^i$ ensures that agents always find it preferable to repay their debts. The scenario is now equivalent to Problem 4.2. Equilibria are not Pareto efficient because the market constraint depends on prices.

Chapter 5

Sequential Trading Economies

5.1 Incomplete Financial Market with $L = 1$

5.1.1 CPE when $L = 1$

.

(Diamond) A financial market equilibrium $(x, \pi) \in \mathbb{R}_+^{(S+1)I} \times \mathbb{R}_{++}^S$ of an economy with $\text{rank}(A) = J < S$ is constrained Pareto Efficient when $L = 1$.

Proof. We prove by contradiction, as the proof of the First Welfare Theorem. When $L = 1$, the spanning condition becomes

$$\begin{bmatrix} \vdots \\ y_s^i - w_s^i \\ \vdots \end{bmatrix} \in \text{span}(A)$$

Let x be a financial market equilibrium. Suppose there exists a y that is preferred by all i (with strict preference for at least one j .) Then, we have

$$\begin{aligned} y_0^i - w_0^i + \sum_s \pi_s (y_s^i - w_s^i) &\geq x_0^i - w_0^i + \sum_s \pi_s (x_s^i - w_s^i) \\ y_0^j - w_0^j + \sum_s \pi_s (y_s^j - w_s^j) &> x_0^j - w_0^j + \sum_s \pi_s (x_s^j - w_s^j) \end{aligned}$$

Summing across all i and invoking the Walras' Law, we have

$$\sum_i y_0^i - w_0^i + \sum_s \pi_s \sum_i (y_s^i - w_s^i) > 0$$

Since $\pi_s > 0$, we have $\sum_i (y_s^i - w_s^i) > 0$, contradicting the feasibility condition.
Q.E.D.

5.2 The Representative Agent Theorem

Statement

Theorem (Representative Agent). Suppose all I agents have identical VNM preferences with the same homothetic Bernoulli utility:

$$U^i(x^i) = u(x_0^i) + \sum_{s=1}^S \text{prob}_s u(x_s^i), \quad u(\cdot) \text{ homothetic, } \forall i.$$

If markets are complete ($\text{rank}(A) = S$), then there exists a representative agent with

$$U^R(x) = U(x), \quad \omega^R = \sum_{i=1}^I \omega^i,$$

such that the FME of the single-agent economy has the same prices (p, q) as the original I -agent FME. The representative agent's equilibrium allocation is autarchic: $x^R = \omega^R$, $z^R = 0$.

Three ingredients (memorize these)

1. **Identical preferences** $\Rightarrow$ same MRS structure across agents.
2. **Homotheticity** $\Rightarrow$ individual demand is linear in wealth: $x^i(p, y^i) = y^i x(p, 1)$, so demand aggregates linearly.
3. **Complete markets** ($\text{rank}(A) = S$) $\Rightarrow$ equilibrium is Pareto efficient and equalizes MRSs across agents, so only aggregate wealth matters.

Proof sketch

Step 1: Linear aggregation from homotheticity. By homotheticity, $x^i(p, y^i) = y^i x(p, 1)$. Identical preferences give $x^i(p, 1) = x(p, 1)$. Summing across agents:

$$\sum_i x^i(p, y^i) = \left(\sum_i y^i \right) x(p, 1) = x \left(p, \sum_i y^i \right).$$

The last equality reapplies homotheticity in reverse — aggregate demand equals the demand of a single agent endowed with $\sum_i y^i$.

Step 2: Walras + spot market clearing. The budget constraint gives $p_s x_s^i = y_s^i$. Summing and using market clearing $\sum_i x_s^i = \sum_i \omega_s^i$:

$$\sum_i y_s^i = p_s \sum_i \omega_s^i.$$

Step 3: Combine. Substituting into the aggregated demand:

$$x\left(p_s, p_s \sum_i \omega_s^i\right) = \sum_i \omega_s^i.$$

This is the equilibrium condition for a *single* agent who owns $\sum_i \omega^i$ and faces prices p . Hence the same p supports both equilibria.

Step 4: Where completeness enters. Complete markets ensure the original I -agent equilibrium is Pareto efficient, so all agents share a common pricing kernel m . This is what makes the aggregation valid *across states* (not just within a state) and pins down q from the representative agent's MRS: $q = \mathbb{E}[m^R A]$.

Why it matters

The theorem reduces equilibrium price computation to a *single-agent* problem: just solve

$$q_j = \sum_s \text{prob}_s \frac{u'(\sum_i \omega_s^i)}{u'(\sum_i \omega_0^i)} a_{sj},$$

plugging in aggregate endowments. No need to solve for the individual allocation $(x^i, z^i)_{i=1}^I$. This is the foundation of essentially all macro/finance asset-pricing models (CCAPM, Lucas tree, etc.).

Memory hooks

- **Slogan:** “Identical + homothetic + complete $\Rightarrow$ aggregate.”
- **Each assumption kills one obstruction:** identical preferences kill heterogeneous demand functions; homotheticity kills wealth-distribution effects; completeness kills MRS dispersion across agents.
- **Compare to uniqueness Proposition 6(ii):** the static version (identical + homothetic) gives uniqueness in a pure exchange economy. The representative agent theorem is the dynamic/state-contingent analogue, with completeness added to handle the extra dimension.
- **Equilibrium is autarchic:** $x^R = \omega^R$, $z^R = 0$ — the representative agent trades with herself, so net trades vanish.

Chapter 6

General Equilibrium with Incomplete Markets (Primer)

6.1 Constrained PE in production economies with incomplete markets ($L = 1$)

6.1.1 Takeaway

Consider a financial market economy that is already Constrained Pareto efficient, adding production and equity trades

- does NOT induce inefficiency if firms adopt Minkowski's rational price conjecture;
- induces inefficiency under Dr  rez price conjecture.

6.2 Pecuniary externalities and constrained inefficiency

In this section, we introduce several examples of the so-called “Pecuniary Externalities”, where the prices have dual roles in the economy:

- They clear the markets;
- They also affect the restrictions on agents' choice set. Concretely, the consumption allocation is restricted by:

$$\begin{bmatrix} \vdots \\ x_s^i \\ \vdots \end{bmatrix} \in B^i(p_1, \dots, p_S).$$

An important proposition that relates to the second role of prices in the GEI is the following:

Proposition 8. *Constrained Pareto Efficiency $\Rightarrow B^i \perp p$.*

Equivalently, If B^i depends on prices p , the equilibrium allocation is generally Constrained Inefficient.^a

^aAn exception is homothetic preferences.

The converse is not true for the proposition above if $L > 1$. But by the Diamond's theorem, we have if $L = 1$ (single good economy), then

Proposition 9. $B^i \perp p \Leftrightarrow \text{Constrained Pareto Efficiency}$ when $L = 1$.

Consider an economy with agents $i \in \mathcal{I}$, two dates $t = 0, 1$, and states $s \in \mathcal{S}$ at date 1 with probabilities π_s . Let $z \in R^K$ be the asset portfolio vector with date-0 price vector q . The asset payoff in state s is $a_s \in R^K$.

Consumer i 's Problem:

$$\begin{aligned} \max_{\{x^i, z^i\}} \quad & u^i(x_0^i) + \sum_{s \in \mathcal{S}} \pi_s u^i(x_s^i) \\ \text{s.t.} \quad & p_0(x_0^i - w_0^i) + q \cdot z^i = 0 \\ & p_s(x_s^i - w_s^i) = a_s \cdot z^i, \quad \forall s \in \mathcal{S} \end{aligned}$$

6.3 Default

6.3.1 Environment

Consider an economy with I agents, S states, a single commodity ($L = 1$), and a full set of Arrow securities: $A = I_S$ (where I_S is an S -dimensional identity matrix.) Agents can default in each state $s \in \mathcal{S}$, after the state is realized. If they default they consume only a fraction $\alpha \in (0, 1)$ of their endowment. The remaining fraction, $1 - \alpha$, is first pooled across all defaulting agents and then redistributed pro-rata to their creditors.

6.3.2 Equilibrium

Definition: A competitive equilibrium with default under complete financial market is defined by asset prices ($q \in \mathbb{R}_{++}^S$), consumption choices $x^i \in \mathbb{R}_{++}$, and default choices $d^i \in \{0, 1\}$, such that

1. Each consumer solves

$$\begin{aligned} & \max_{x_0^i, x_s^i, d^i} u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i) \\ \text{subject to: } & x_0^i - w_0^i + qz^i = 0 \\ & \forall s \in S, \quad x_s^i = \begin{cases} w_s^i + \underbrace{\frac{z_s^i}{\sum_i \max\{0, z_s^i\}}}_{\text{pro-rata weight}} \sum_{j \neq i} \left[d_j(1 - \alpha)w_s^j + (1 - d_j) \underbrace{\max\{0, -z_s^j\}}_{\text{identifying borrowers}} \right], & \text{if } z_s^i \geq 0 \\ d_i \alpha w_s^i + (1 - d_i)(z_s^i + w_s^i), & \text{if } z_s^i < 0 \quad (i \text{ is a borrower}) \end{cases} \end{aligned}$$

where I use the state index s to denote the asset as the financial market is complete.

2. Goods market clears

$$\sum_i x^i = \sum_i w^i$$

3. Asset market clears

$$\sum_i z^i = 0$$

6.3.3 Efficiency

Consider a situation where $J = S - 1$, $L = 1$, and no-default constraints are imposed on assets $j = 2, 3, \dots, S - 1$.

- Is competitive equilibrium constrained Pareto Efficient?

Note that the no-default condition is

$$w_s^i + z_s^i > \alpha w_s^i$$

$\Rightarrow$

$$z_s^i > -(1 - \alpha)w_s^i$$

Now consider the budget constraint:

$$\begin{aligned} x_0^i - w_0^i + \sum_s q_s z_s^i &= 0 \\ x_s^i - w_s^i &= z_s^i \end{aligned}$$

Using No-Arbitrage condition, we have $q = \pi A$, As a result, the budget constraint becomes

$$x_0^i - w_0^i + \sum_s \pi_s (x_s^i - w_s^i) = 0 \quad \text{and} \quad \underbrace{\begin{bmatrix} \vdots \\ x_s^i - w_s^i \\ \vdots \end{bmatrix}}_{\text{Doesn't depend on prices}} \in \mathbb{R}^{S-1} \cap (\text{No-Default constrained set})$$

Since $L = 1$, the constained set doesn't depend on prices of goods or assets, the competitive equilibrium is **constrained Pareto Efficient**.

Chapter 7

Sequential Production Economies

Timeline

Firms buy at time 0 on the spot market and sell at time 1 on the state-contingent markets.

7.1 Main concepts in this section

- Arrow-Debreu equilibrium
- Financial Market equilibrium
- Minkowski's rational price conjecture and Dreze conjecture
- The most important proof: ($L = 1$, $H = 1$, $J = 0$)
Competitive equilibria of FME with production under Rational Conjectures are Constrained P.E.
- Corporate finance (e.g., equity vs. bonds)

Notations

- δ^{ih} : endowment of firms' shares
- ϕ : prices
- θ : quantity of equity trade

7.2 Arrow-Debreu Equilibrium (with Production)

Definition.

An Arrow-Debreu Equilibrium is $(x, y, \phi) \in \mathbb{R}_+^{L(S+1)I} \times \Upsilon \times \mathbb{R}_{++}^{L(S+1)}$ (N.B.: $y = [y^h]$, with $y^h h = (-k^h, f^h(k^h))$.) such that

1. Given ϕ , each individual solves the utility maximization pb:

$$\begin{aligned} \max_{x_0^i, x_s^i} \quad & u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i) \\ \text{subject to:} \quad & \phi_0(x_0^i - w_0^i) + \sum_s \phi_s(x_s^i - w_s^i) - \sum_h \delta^{ih} \pi^h = 0 \end{aligned}$$

where π^h is the expected profit of firm h .

2. Each firm solves the following pb:

$$\max_{k^h \in K \subseteq \mathbb{R}_+^L} -\phi_0 k^h + \sum_s \phi_s f^h(k^h; s)$$

3. Market clears¹

$$\sum_i x^i \leq \sum_i w^i + \sum_h y^h$$

where $x^i \in \mathbb{R}_+^{L(S+1)}$.

4. Profits are

$$\pi^h = -\phi_0 k^h + \sum_s \phi_s f^h(k^h; s)$$

¹Since $y^h = [-k^h, f^h(k^h)]$, this condition is the same as

$$\begin{aligned} \sum_i x_0^i &\leq \sum_i w_0^i - \sum_h k^h \\ \sum_i x_s^i &\leq \sum_i w_s^i + \sum_h f^h(k^h; s) \end{aligned}$$

7.3 Financial Market Equilibrium (with Production)

Full Model	Simplified ($J = 0, L = H = 1$)
Consumer solves $\max_{x^i, \theta^i, z^i} u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i)$ s.t. $p_0(x_0^i - w_0^i) + qz^i = (-p_0k + Q)\theta_0^i - Q\theta^i$ $p_s(x_s^i - w_s^i) = \theta^i p_s f^h(k; s) + a_s z^i$	Consumer solves $\max_{x^i, \theta^i, z^i} u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i)$ s.t. $(x_0^i - w_0^i) = (-k + Q)\theta_0^i - Q\theta^i$ $(x_s^i - w_s^i) = \theta^i p_s f^h(k; s)$
Firm solves $\max_{k^h} -p_0 k^h + Q(k^h)$	Firm solves $\max_{k^h} -k^h + Q(k^h)$
Market clears $\sum_i x_0^i + \sum_h k^h \leq \sum_i w_0^i$ $\sum_i x_s^i \leq \sum_i w_s^i + \sum_h f^h(k^h; s)$ $\sum_i z^i = 0 \quad \text{and} \quad \sum_i \theta^i = 1$	Market clears $\sum_i x_0^i + \sum_h k^h \leq \sum_i w_0^i$ $\sum_i x_s^i \leq \sum_i w_s^i + \sum_h f^h(k^h; s)$ $\sum_i \theta^i = 1$
Rational Conjectures $Q^h(k^h) = \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$	Rational Conjectures $Q^h(k^h) = \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$
Consistency in equity prices $Q^h = Q^h(k^h), \quad \forall h \in H$	Consistency in equity prices $Q^h = Q^h(k^h), \quad \forall h \in H$

Proposition 10. *Let (θ, k) be an equilibrium allocation. Consider an enlarged economy where agents can hold equity θ_e^i in the original firm $f(k)$ and γ_e^i in an off-equilibrium firm $f(k')$. Then:*

- *Under **Rational Conjectures**, $\gamma_e^i = 0$ for all i .*
- *Under **Drèze Conjectures**, the condition $\gamma_e^i = 0$ for all i is not guaranteed to hold (trade may occur).*

Proof (Sketch):

Mathematically, this result is because

$$\begin{aligned} \mathbb{E}[m^i p \circ f^h(k^{h'})] &\leq Q^h(k^{h'}), \quad \forall i \\ \exists j, \text{ such that } \mathbb{E}[m^j p \circ f^h(k^{h'})] &> D^h(k^{h'}) \end{aligned}$$

In words, the proposition says:

- In the case where the equity is priced using the highest-bidder's evaluation, no one is going to buy the off-equilibrium firms if the highest-bidder doesn't buy it.
- However, if the equity of the off-equilibrium firms are priced using equilibrium holders' values, there might exist a potential buyer who values these firms more and thus want to buy them.

Proposition 11. *Assuming $J = 0, L = H = 1$, competitive equilibria of the FME with production are constrained P.E.*

The key of the proof relies on the optimization of the firm.

Proof (Sketch): Let (x, θ, k) be a competitive equilibrium. Suppose there exists $(x_0^{i'}, \theta^{i'}) \succeq_P (x_0^i, \theta^i)$, then by the Law of Revealed Preference, we have

$$x_0^{i'} + Q(k')\theta^{i'} \geq x_0^i + Q(k)\theta^i \forall i$$

where some i the inequality is strict. Now using the budget constraint and the consistency in the eqm, $(x_0^i = w_0^i + (-k + Q(k))\theta_0^i - Q(k)\theta^i)$, we have

$$\begin{aligned} x_0^{i'} + Q(k')\theta^{i'} &\geq x_0^i + Q(k)\theta^i \\ &= w_0^i + (-k + Q(k))\theta_0^i - Q(k)\theta^i + Q(k)\theta^i \\ &= w_0^i + (-k + Q(k))\theta_0^i \end{aligned}$$

Summing over i , and applying the market clearing condition $(\sum_i \theta^{i'} = 1)$, then we have the following strict inequality

$$Q(k') + \sum_i x_0^{i'} > -k + Q(k) + \sum_i w_0^i$$

Now here comes the critical step: Due to the optimization of the firm, we have

$$-k + Q(k) \geq -k' + Q(k')$$

It follows that

$$Q(k') + \sum_i x_0^{i'} > -k' + Q(k') + \sum_i w_0^i$$

Hence, we derive a contradiction: $\sum_i x_0^{i'} + k' > \sum_i w_0^i$. ■

7.4 Rational price conjectures

Proposition 12. Define $Q^h(k^h) := \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$, and $D^h(k^h) := \mathbb{E}[\sum_i \theta_h^i m^i p \circ f^h(k^h)]$, we have

$$FME \text{ with Rational Conjectures } (Q^h) \quad \not\Rightarrow \quad FME \text{ with Dreze Conjectures } (D^h)$$

Proof (Sketch): Note that in equilibrium, these equity prices coincide with RC and Dreze Conjecture, namely, $Q^h(k^{h*}) = D^h(k^{h*})$, where k^{h*} is the equilibrium input choice of firm h . However, away from equilibrium, we have

$$Q^h(k^h) \geq D^h(k^h)$$

Hence, there are no profitable deviations from equilibrium under RC, (that is the $\Rightarrow$ part), but there could be profitable deviations under Dreze (that is the $\not\Rightarrow$ part). ■

7.5 Corporate Finance

7.5.1 FME with bonds and equity

Definition. A competitive equilibrium of a Financial Market Economy with production, rational price conjecture, under 2 periods (assuming $I > 1$ individuals, S states, $J = 1$ assets, and $H = 1$ firm) is defined by prices and quantities $(x, z, \theta, B, k, q, Q, P) \in \mathbb{R}_+^{L(S+1)I} \times \mathbb{R}_+^{JI} \times \mathbb{R}_+^{HI} \times \mathbb{R}_+^{HI} \times \mathbb{R}_+^{LH} \times \mathbb{R}_+^J \times \mathbb{R}_+^H \times \mathbb{R}_+^H$, such that

1. Each consumer solves

$$\begin{aligned} \max_{x^i, z^i, \theta^i, B^i} \quad & u^i(x_0^i) + \sum_{s \in S} \text{prob}_s u^i(x_s^i) \\ \text{subject to:} \quad & x_0^i - \omega_0^i = (-k + Q + PB) \theta_0^i - Q \theta^i - PB \theta^i - q z^i \\ & x_s^i - \omega_s^i = a_s z^i + \theta^i (f(k^h; s) + B), \forall s \in S \end{aligned}$$

2. Each firm solves

$$\max_{k^h, B^h} \quad -k^h + Q(k^h, B^h) + P(k^h, B^h) B^h$$

3. Markets clear

$$\begin{aligned}\text{Goods:} \quad & \sum_i (x^i - w^i) = 0 \\ \text{Assets:} \quad & \sum_i z^i = 0 \\ \text{Equity:} \quad & \sum_i \theta^i = 1\end{aligned}$$

4. The price of equity satisfies rational conjectures (for in- and off- equilibrium paths)

$$\begin{aligned}Q^h(k^h, B^h) &= \max_i \mathbb{E} \left[m^i [\max\{0, f(k; s) - B\}] \right], \quad \forall h \\ P^h(k^h, B^h) &= \max_i \mathbb{E} \left[m^i [\min\{1, \frac{f(k; s)}{B}\}] \right], \quad \forall h\end{aligned}$$

5. Consistency: $Q^h = Q^h(k^h, B^h)$ and $P^h = P^h(k^h, B^h)$.

7.5.2 Modigliani-Miller Theorem.

Theorem. If the financial market is complete, the decision of B is indeterminate.

Proof. Under rational conjecture, we have

$$\begin{aligned}Q(k, B) &= \max_i \mathbb{E}[m_i \max\{0, f(k, s) - B\}] \\ P(k, B) &= \max_i \mathbb{E}[m_i \min\{1, \frac{f(k, s)}{B}\}]\end{aligned}$$

Because market is complete, $m^i = m$, hence we have

$$\begin{aligned}Q(k, B) + P(k, B)B &= \mathbb{E}[m \max\{0, f(k, s) - B\} + m \min\{B, f(k, s)\}] \\ &= \mathbb{E}[m \max\{B, f(k, s)\} + m \min\{B, f(k, s)\} - mB] \\ &= \mathbb{E}[mf(k, s)]\end{aligned}$$

which is independent of B .

7.5.3 Modigliani-Miller Theorem when $J < S$ but $\exists$ a common holder

Now we consider the situation where the financial market is incomplete, namely, $\text{rank}(A) = J < S$. However, if there is a common holder, namely, $\arg\max_i Q(m_i) = \arg\max_i P(m_i)$, the Modigliani-

Miller Theorem also holds!

$$Q(k, B) + P(k, B)B = \mathbb{E}[m_{i^*} f(k, s)]$$

where i^* is the highest bidder who holds both equity and bond.

7.6 Problems

7.6.1 Problem 5.1

Problem. Consider a definition of Competitive equilibrium which differs in the specification of a consistency condition on price conjectures — which we now denote with $D^h(k^h)$. Consider in particular the price conjectures proposed by Dreze (1974). Let θ_h^i be the position of agent i on the equity of firm h . Then, for any $h \in H$, let the Dreze conjectures be defined as:

$$D^h(k^h) = \mathbb{E} \left[\sum_{i \in I} \theta_h^i m^i p f^h(k^h) \right].$$

Note that these conjectures only depend on the marginal valuation of agents $i \in I$ such that $\theta_h^i > 0$. Show that an equilibrium with Rational Conjectures is also an equilibrium with Dreze conjectures but the converse does not hold.

Solution. Suppose $\theta_h^i > 0$. If $\mathbb{E}[m^i p \square f^h(k_*^h)] < Q^h(k_*^h)$, then agent i would like to decrease θ_h^i . Thus, in any equilibrium, $\theta_h^i > 0$ iff $\mathbb{E}[m^i p \square f^h(k_*^h)] = Q^h(k_*^h)$. Therefore, $Q^h(k_*^h) = D^h(k_*^h)$.

Start from a RC equilibrium. There are no profitable deviations under Q^h . Substitute Q^h with D^h . Since $D^h(k_*^h) = Q^h(k_*^h)$, consistency holds. Since $D^h \leq Q^h$, there are no profitable deviations under D^h : Dreze eq.

Start from a Dreze equilibrium. There are no profitable deviations under D^h . Substitute D^h with Q^h . Since $Q^h(k_*^h) = D^h(k_*^h)$, consistency holds. Since $Q^h \geq D^h$, there might be profitable deviations under Q^h : no RC eq.

7.6.2 Problem 5.2

Problem. Identify precisely how the proof of Lemma 5.1 (which shows that enlarging the asset space with an off-equilibrium firm leaves the equilibrium unchanged under rational conjectures) breaks down if you use e.g. Dreze conjectures.

Solution. With Dreze conjectures, it no longer holds that $\mathbb{E}[m^i p \square f^h(k')] \leq D^h(k')$ for all i , so some agent may be willing to buy shares of the off-equilibrium firm.

7.6.3 Problem 5.3

Problem. Consider an economy with $L = 1$ and — for simplicity but without loss of generality — $H = 1$. Identify precisely how the proof of Lemma 5.1 breaks down if we allow for positive finite short sales $\bar{\theta} > 0$.

Solution. With short sales and rational conjectures, every agent with $\mathbb{E}[m^i p \square f(k')] < Q(k')$ finds it preferable to sell shares of the off-equilibrium firm. Therefore, $\gamma_e = 0$ is incompatible with eq of the augmented economy.

Chapter 8

Assignment and Search

Proposition 13. *The dual assignment and the primal assignment problems yield the same solution.*

Proof The Primal problem is given by

$$\begin{aligned} \max_{\pi^{ih}} \quad & \sum_i \sum_h \pi^{ih} U^{ih} \\ \text{s.t.} \quad & \sum_h \pi^{ih} = w^i, \quad \forall i \\ & \sum_i \pi^{ih} = f^h, \quad \forall h \end{aligned}$$

The Dual problem is given by

$$\begin{aligned} \min_{w^i, f^h} \quad & \sum_i u^i w^i + \sum_h v^h f^h \\ \text{s.t.} \quad & u^i + v^h \geq U^{ih}, \quad \forall i, h \end{aligned}$$

The Lagrangean for the Primal problem is

$$\mathcal{L}^P = \sum_i \sum_h \pi^{ih} U^{ih} + \sum_h v^h \left(f^h - \sum_i \pi^{ih} \right) + \sum_i u^i \left(w^i - \sum_h \pi^{ih} \right)$$

The Lagrangean for the Dual problem is

$$\mathcal{L}^D = \sum_i u^i w^i + \sum_h v^h f^h + \sum_i \sum_h \pi^{ih} \left(U^{ih} - u^i - v^h \right)$$

Obviously, when choosing proper Lagrangean Multipliers, namely, using $\{v^h\}, \{u^i\}$ as the LM for the Primal problem and $\{\pi^{ih}\}$ for the Dual problem, these two Lagrangean coincide with each other and thus yield the same solutions. ■

Proposition 14. *If U^{ih} is supermodular, then the optimal matching is assortative.*

Proof Mathematically, it is a direct implication of the Hardy-Littlewood-Polya inequality, which says

$$\langle x, y \rangle \leq \langle \hat{x}, \hat{y} \rangle,$$

where $\hat{x}, \hat{y}$ are the decreasing rearrangements of x, y respectively.

Now, we prove the proposition by contradiction with inductive steps. Assume that the optimal matching is not assortative; concretely, assume that worker i matches with firm $\sigma_*(i) \neq i$. Then, using the supermodularity of U , we have

$$\sum_{i \neq 1 \neq \sigma_*^{-1}(1)} U^{i\sigma_*(i)} + U^{11} + U^{\sigma_*^{-1}(1)\sigma_*(1)} > \sum_i U^{i\sigma_*(i)}.$$

In the k^{th} step, we have

$$\sum_{i \neq k \neq \sigma_*^{-1}(k)} U^{i\sigma_*(i)} + U^{kk} + U^{\sigma_*^{-1}(k)\sigma_*(k)} > \sum_i U^{i\sigma_*(i)}.$$

Iterate this procedure for all i , we reach a contradiction. ■

8.1 Problems

8.1.1 Problem 3.1

Problem. Prove Proposition 3.2 (equivalence between stability of (π, u, v) and the property that π solves the primal assignment problem while (u, v) solves its dual).

Solution. “If” side. Suppose that π is a solution to the primal problem and (u, v) is a solution to the dual problem. By Lemma 2:

$$U^{ih} = u^i + v^h.$$

Since π solves the primal problem:

$$\sum_{i \in I} \sum_{h \in H} \pi^{ih} U^{ih} = \sum_{i \in I} \sum_{h \in H} \pi^{ih} u^i + \sum_{h \in H} \sum_{i \in I} \pi^{ih} v^h = \sum_{i \in I} w^i u^i + \sum_{h \in H} f^h v^h.$$

Since (u, v) solves the dual problem, for all $i \in I$ and $h \in H$ we have $u^i + v^h \geq U^{ih}$. Therefore, (π, u, v) is a stable tuple.

“Only if” side. Suppose that (π_*, u_*, v_*) is a stable tuple. By absence of blocking pairs, for all $\pi \in \Pi$ we have:

$$\sum_{i \in I} \sum_{h \in H} \pi^{ih} U^{ih} - \sum_{i \in I} \sum_{h \in H} \pi^{ih} u_*^i - \sum_{h \in H} \sum_{i \in I} \pi^{ih} v_*^h \leq 0.$$

Therefore:

$$\max_{\pi \in \Pi} \sum_{i \in I} \sum_{h \in H} \pi^{ih} U^{ih} \leq \sum_{i \in I} w^i u_*^i + \sum_{h \in H} f^h v_*^h \quad \text{s.t.} \quad \sum_{h \in H} \pi^{ih} = w^i, \quad \sum_{i \in I} \pi^{ih} = f^h.$$

By feasibility:

$$\sum_{i \in I} w^i u_*^i + \sum_{h \in H} f^h v_*^h = \sum_{i \in I} \sum_{h \in H} \pi_*^{ih} U^{ih} \leq \max_{\pi \in \Pi} \sum_{i \in I} \sum_{h \in H} \pi^{ih} U^{ih}.$$

Therefore, π_* solves the primal problem. Moreover,

$$\begin{aligned} \min_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^H} \max_{\pi \in \Pi} S(\pi, u, v) &\leq \max_{\pi \in \Pi} S(\pi, u_*, v_*) \\ &= S(\pi_*, u_*, v_*) = \sum_{i \in I} w^i u_*^i + \sum_{h \in H} f^h v_*^h \\ &= \sum_{i \in I} \sum_{h \in H} \pi_*^{ih} U^{ih} = \max_{\pi \in \Pi} \min_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^H} S(\pi, u, v) \\ &= \min_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^H} \max_{\pi \in \Pi} S(\pi, u, v). \end{aligned}$$

Therefore, (u_*, v_*) solves the dual problem.

8.1.2 Problem 3.2

Problem. Show that, if $I = H$ and U is supermodular, the solution to the Monge problem is the identity permutation; that is, each agent/worker $i \in I$ is paired with firm $h = i$.

Solution. We prove by induction that an optimal solution σ_* must be such that $\sigma_*(i) = i$ for all $i \in I$.

- *Initial step.* Suppose that σ_* is optimal but $\sigma_*(1) \neq 1$. Then:

$$\sum_{1 \neq i \neq \sigma_*^{-1}(1)} U^{i\sigma_*(i)} + U^{11} + U^{\sigma_*^{-1}(1)\sigma_*(1)} > \sum_{i \in I} U^{i\sigma_*(i)}.$$

Therefore, σ_* is not an optimal solution.

- *Inductive hypothesis.* If σ_* is an optimal solution, then $\sigma_*(i) = i$ for all $i < k$.
- *Inductive step.* Suppose that σ_* is optimal but $\sigma_*(k) \neq k$. By inductive hypothesis, $\sigma_*(k) > k$ and $\sigma_*^{-1}(k) > k$. But then:

$$\sum_{k \neq i \neq \sigma_*^{-1}(k)} U^{i\sigma_*(i)} + U^{kk} + U^{\sigma_*^{-1}(k)\sigma_*(k)} > \sum_{i \in I} U^{i\sigma_*(i)}.$$

Therefore, σ_* is not an optimal solution.

Chapter 9

Asymmetric Information

9.1 Prescott-Townsend Competitive Equilibrium

Definition. A competitive equilibrium in a Moral Hazard Economy is defined by an allocation $(x, e) \in \mathbb{R}^S \times E$, and prices $q \in \mathbb{R}^{SE}$ such that

1. Each consumer i solves

$$\begin{aligned} & \max_{x^i, e} \quad \sum_s \text{Prob}_s(e) u(x_s) - v(e) \\ \text{subject to:} \quad & \sum_s q_s(e)(x_s - w_s) = 0 \end{aligned}$$

2. Market clears

$$\sum_s \text{Prob}_s(e)(x_s^i - w_s^i) = 0,$$

where due to the Law of Large Number, $\text{Prob}_s(e)$ is the fraction of consumers choose effort e when state s is realized.

3. Self-imposing incentive constraint:

$$e \in \text{argmax}_{\epsilon} \sum_s \text{Prob}_s(\epsilon) u(x_s) - v(\epsilon), \quad \text{given } x$$

4. Rational conjecture

$$\text{Prob}_s(e) = q_s(e)$$

9.2 Radner Rational Expectation Equilibrium

For an economy with $t = 0, 1$, S states, J assets, $L = 1$ commodity, $I > 1$ agents with strictly positive endowments and strictly monotonic, vNM, and strictly quasiconcave preferences, and Σ^I signals. A Radner Rational Expectation Equilibrium consists of $(x, z, q_\sigma, \phi) \in \mathbb{R}_+^{(S+1)I} \times \mathbb{R}_+^{JI} \times \mathbb{R}_{++}^{J|\Sigma|^I} \times \Phi$ such that

1. Given $w_i, A, q_\sigma, \phi, \sigma_i$, agent i solves the following problem

$$\begin{aligned} \max_{x_i, q_i} & U_i(x_0^i) + \sum_s \text{prob}_s^{\sigma_i, \phi^{-1}(q_\sigma)} U_i(x_s^i) \\ \text{subject to: } & x_0^i - w_0^i + q_\sigma z_i = 0 \\ & x_s^i - w_s^i = a_s z_i, \quad \forall s \end{aligned}$$

where $\text{prob}_s^{\sigma_i, \phi^{-1}(q_\sigma)}$ is the posterior beliefs about the state s given i 's signal σ_i and her information about others' signals given the inverse price-map and the price $\phi^{-1}(q_\sigma)$.

2. Markets clear

- Assets (signal-dependent): $\sum_i z_i(q_\sigma, \phi) = 0, \forall \sigma \in \Sigma^I$
- Goods at $t = 0$: $\sum_i x_0^i - w_0^i = 0$
- Goods at $t = 1$: $\sum_i x_s^i - w_s^i = 0, \forall s$

3. Assets prices are consistent: $q_\sigma = \phi(\sigma)$.

9.2.1 Fully Revealing Equilibria

Definition 9.2.1. A REE price map $\phi : \Sigma^I \rightarrow \mathbb{R}_+^J$ is *fully revealing* if

$$\phi(\sigma) \neq \phi(\sigma'), \quad \text{for any } \sigma, \sigma' \in \Sigma^I \text{ with } \sigma \neq \sigma'.$$

Equivalently, ϕ is invertible: by observing q_σ , every agent can back out the joint signal profile $\sigma \in \Sigma^I$, so the posterior conditional on $(\sigma^i, \phi^{-1}(q_\sigma))$ collapses to the full-information posterior conditional on σ ,

$$\text{prob}_s^{\sigma^i, \phi^{-1}(q_\sigma)} = \text{prob}_s^\sigma.$$

Theorem 2 (Radner 1979). *For a generic subset of economies parametrized by endowments $\omega \in \mathbb{R}_{++}^{(S+1)I}$, there exists a REE price map $\phi : \Sigma^I \rightarrow \mathbb{R}_+^J$ that is fully revealing.*

Proof sketch (counting equations and unknowns). Because a fully revealing ϕ is invertible, every agent's posterior is the full-information posterior prob_s^σ . Consider the auxiliary economy

in which agent i 's problem is

$$\begin{aligned} \max_{(x^i, z^i) \in \mathbb{R}_+^{S+1} \times \mathbb{R}^J} \quad & u^i(x_0^i) + \sum_{s \in S} \text{prob}_s^\sigma u^i(x_s^i) \\ \text{subject to:} \quad & x_0^i + q_\sigma z^i \leq \omega_0^i, \\ & x_s^i \leq a_s z^i + \omega_s^i, \quad \forall s. \end{aligned}$$

Let $z^i(q_\sigma, \phi)$ denote the optimal portfolio and $Z(q_\sigma, \phi) = \sum_{i \in I} z^i(q_\sigma, \phi)$ the aggregate excess demand for the J assets.

We argue by contrapositive: suppose ϕ is *not* fully revealing, i.e., there exist $\sigma \neq \sigma' \in \Sigma^I$ such that $\phi(\sigma) = \phi(\sigma')$. Then the following three conditions must hold simultaneously:

$$\begin{aligned} Z(\phi(\sigma), \phi) &= 0, \\ Z(\phi(\sigma'), \phi) &= 0, \\ \phi(\sigma) &= \phi(\sigma'). \end{aligned}$$

Count: treating $\phi(\sigma), \phi(\sigma') \in \mathbb{R}^J$ as the unknowns, this is a system of $3J$ equations in $2J$ unknowns. By the Transversality Theorem applied to the endowment parameter $\omega \in \mathbb{R}_{++}^{(S+1)I}$, an over-determined system of this form generically (off a closed, measure-zero subset of ω) has no solution. Hence generically no two distinct signal profiles deliver the same equilibrium price, and the REE is fully revealing. $\square$

Remarks.

- At a fully revealing REE, all agents share the same posterior prob_s^σ , so the equilibrium allocation is exactly the one in a full-information financial market economy: *agents do not trade on account of information*, only on account of risk-sharing.
- The genericity logic is the same as the one used in incomplete markets (see Geanakoplos-Polemarchakis): a transversality count $3J$ vs. $2J$ pins down the result, even though we have not made the perturbation argument explicit.
- The result *fails* when the signal space is too rich relative to the asset span. If $\sigma \in \mathbb{R}^{\Sigma I}$ is continuous and $\Sigma I > J$, the price map $\phi : \mathbb{R}^{\Sigma I} \rightarrow \mathbb{R}^J$ generically cannot be invertible by a dimensionality argument: ϕ maps a ΣI -dimensional set into a J -dimensional one (Theorem 6.2 / Allen 1981).

9.3 Moral Hazard in Production

9.3.1 Static

9.4 Problems

9.4.1 Problem 6.1

Problem. Characterize Incentive constrained optima and competitive equilibria (with the consistency condition on price maps) of the moral hazard economy specialized so that

$$S = \{1, 2\}; \quad e \in \{h, \ell\}; \quad v(h) > v(\ell); \quad \pi_1(h) > \pi_1(\ell); \quad \omega_1 > \omega_2.$$

Set-up and characterize equilibria of this economy under the constraint that prices are linear (independent of x), $q_s(e(x)) = q_s$. Are equilibrium allocations in this case Incentive constraint efficient?

Solution. An IC optimum (x^*, e^*) solves:

$$\begin{aligned} & \max_{x, e} \pi_1(e)u(x_1) + (1 - \pi_1(e))u(x_2) - v(e) \\ \text{s.t.} \quad & \begin{cases} \pi_1(e)(x_1 - \omega_1) + (1 - \pi_1(e))(x_2 - \omega_2) = 0 \\ e \in \arg \max_{\varepsilon} \pi_1(\varepsilon)u(x_1) + (1 - \pi_1(\varepsilon))u(x_2) - v(\varepsilon) \end{cases} \end{aligned}$$

A competitive equilibrium (x^*, e^*, q) clears markets and solves:

$$\begin{aligned} & \max_{x, e} \pi_1(e)u(x_1) + (1 - \pi_1(e))u(x_2) - v(e) \\ \text{s.t.} \quad & \begin{cases} q_1(x_1 - \omega_1) + q_2(x_2 - \omega_2) = 0 \\ e \in \arg \max_{\varepsilon} \pi_1(\varepsilon)u(x_1) + (1 - \pi_1(\varepsilon))u(x_2) - v(\varepsilon) \end{cases} \end{aligned}$$

Moreover, $q_s = \pi_s(e^*)$. Equilibria are in general not efficient. High effort might be supported in the IC optimum if $\pi_1(h)$ is sufficiently larger than $\pi_1(\ell)$ and ω_1 is sufficiently larger than ω_2 . However, high effort cannot be supported in equilibrium because prices do not change with the choice of x .

9.4.2 Problem 6.2

Problem. Write the moral hazard economy with idiosyncratic uncertainty, as in this section, but allowing for consumption at time $t = 0$. Be careful in writing the consistency condition on prices and compare it with Rational conjectures as we formulated them in the production economies with Incomplete markets.

Solution. A competitive equilibrium (x^*, z^*, e^*, q) clears markets and solves:

$$\begin{aligned} & \max_{x, e, z} u(x_0) + \sum_{s \in S} \pi_s(e) u(x_s) \\ \text{s.t. } & \begin{cases} x_0 - \omega_0 + q_0(x)z \leq 0 \\ q_s(x)(x_s - \omega_s) \leq a_s z \\ e \in e(x) = \arg \max_{\varepsilon} u(x_0) + \sum_{s \in S} \pi_s(\varepsilon) u(x_s) \end{cases} \end{aligned}$$

Moreover, $q_s(x) = \pi_s(e(x))$ and $q_0(x) = \sum_{s \in S} \pi_s(e(x)) m_s(x) a_s$ for all x .

Chapter 10

Infinite Horizon Economies

10.1 Bubbles

10.1.1 Why is there no bubbles when T is finite?

[Answer] Intuitively, when T is finite, the last asset holder must equal the price to its fundamental value, which eliminates the possibility of bubbles. Iterating backward, we have no bubble for each period. Formally, we can express asset prices according to

$$p_t = f_t + \mathbb{E}[m_{t,T}d_{t,T}]$$

when $t = T$, the second term equals to 0. Hence, there is no bubble when T is finite.

Chapter 11

Math Preliminaries

11.1 Measure Theory

Definition 11.1.1 (Genericity). A property is considered generic (or holds generically) if it applies to a full measure Lebesgue subset of the space of economies. Consider an economy parameterized by $\omega \in \mathbb{R}_{++}^{LI}$. Intuitively, this means that if you were to randomly select an economy (defined by its endowment vector ω), the property would hold with probability 1. Economies that fail to satisfy this property are “rare” (measure zero) and disappear under slight perturbations.

11.2 Differential Topology

Definition 11.2.1 (Transversality). Let $f : \mathbb{R}^M \rightarrow \mathbb{R}^N$, with $M > N$. We say f is **transversal** at 0 (denoted as $f \pitchfork 0$), if

$$\forall x, \text{ such that } f(x) = 0$$

, we have the Jacobian matrix

$$Df = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_m} \end{bmatrix}$$

is full rank, namely, $\text{rank}(Df) = N$.

Example Consider $M = 2$, and $N = 1$. Define

$$f(x, y) = x^2 + y^2 - 1$$

The zero set is the unit circle. $x^2 + y^2 = 1$.

Since $n = 1$, the Jacobian is a single row vector (the gradient):

$$Df(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \end{bmatrix}$$

The rank of a 1×2 matrix is 1 (full rank) unless all entries are zero.

$$Df = [0, 0] \text{ iff } x = 0 \text{ and } y = 0.$$

However, the origin is **not** in the zero set. Therefore, for every point actually on the circle ($f = 0$), the gradient is non-zero. Hence, we have

$$f \pitchfork 0$$

Example 2: Non-transversal

$$g(x, y) = y^2 - x^3$$

The set of points where $y^2 = x^3$. This describes a semicubical parabola which has a sharp “cusp” (singular point) at the origin.

The Jacobian is given by

$$Dg(x, y) = \begin{bmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} -3x^2 & 2y \end{bmatrix}$$

It follows that $Df(0, 0)$ at the point $(0, 0)$: Therefore, its rank is **0** and thus g is not transversal at 0.

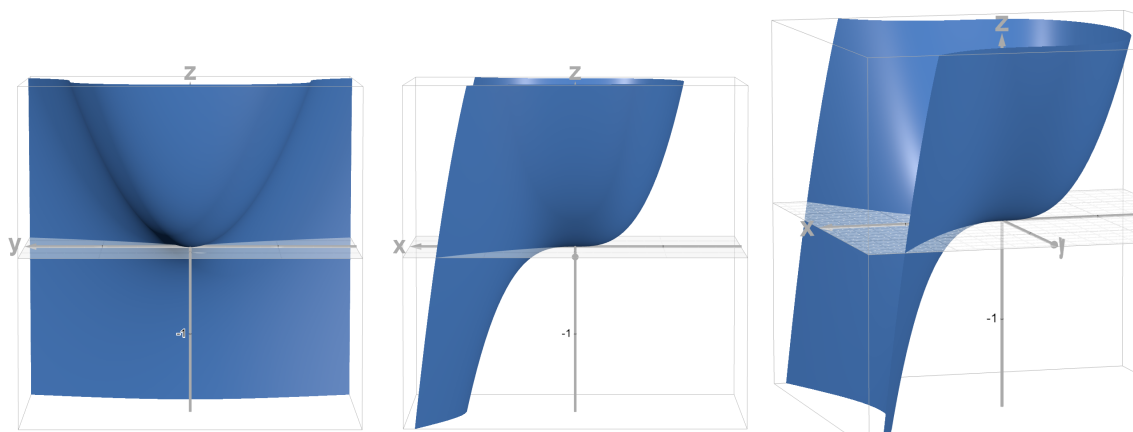


Figure 11.2.1: A function that is not transversal at the origin

NOTES: This function depicts $g(x, y) = y^2 - x^3$, which is not transversal at the origin due to the cusp there.

Proposition 15 (Transversality Theorem). *Let $f : \mathbb{R}^M \rightarrow \mathbb{R}^N$ be C^∞ , with $M > N$. And decompose $x = [x_1, x_2]$, where $x_1 \in \mathbb{R}^{M-N}$, and $x_2 \in \mathbb{R}^N$. For almost every x_1 , if $f(x_1, x_2) = 0$, we*

have

$$\text{rank}(D_{x_2}f) = \begin{bmatrix} \frac{\partial f_1}{\partial x_{2,1}} & \frac{\partial f_1}{\partial x_{2,2}} & \cdots & \frac{\partial f_1}{\partial x_{2,N}} \\ \frac{\partial f_2}{\partial x_{2,1}} & \frac{\partial f_2}{\partial x_{2,2}} & \cdots & \frac{\partial f_2}{\partial x_{2,N}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial x_{2,1}} & \frac{\partial f_N}{\partial x_{2,2}} & \cdots & \frac{\partial f_N}{\partial x_{2,N}} \end{bmatrix} = N.$$

Remarks: We will use transversality theorem to prove a bunch of important results in GE theory, such as the genericity of regular economies.

Chapter 12

Questions:

- What are examples in the literature that show that REE prices might not exist for some economies (the first example is due to D. Kreps)?
- How do we show a production economy with unobserved choices of risk ϕ , $f^h(k^h, \phi)$, presents Moral Hazard?
- Are the following true?
 - With bid-ask spread (i.e., transaction costs), $q \leq \pi A \leq q + \gamma$? where $q = \pi A$ if $\gamma = 0$.
- Why do we have the following statements?
 - (pp. 99) “allowing negative position in firm equity makes small firms change the asset span”
 - In equilibrium, denote the equilibrium input as k^{h*} , then $Q^h(k^{h*}) = D^h(k^{h*})$. In words, prices are the same in equilibrium for both RC and Dreze.
- Confusions about the proofs:
 - I don’t see how Lemma 5.1 is used in the proof of Theorem 5.1.
- How to define a competitive equilibrium with bonds and equity? In particular, how to specify the Bond at time 0?